Standalone Network for ABC Company Branch Office

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Software used: Cisco Packet Tracer

- **Details of the Project**

ABC is a fast growing organization in based in Alabama, with a global customer based exceeding customer base 2 million. The company focuses of buying and selling food items, operating primarily from its Headquarters. To expand its operations, The Company plans to open a new branch near a village of Bolzano. As the part of this ABC has requested to design and implement a standalone network for the Bolzano branch

- **Requirements by ABC Company**

The Branch Network will be a small scale and independent, designed to meet the following criteria

A .Network Devices: One router and one switch to be used

B. Departmental Segmentation: 3 Departments (Admin/IT, Finance/HR and Customer Service/Reception)

C. VLANs Management: Each department is required to be in different VLANs

D. Wireless Connectivity: Each department is required to have a wireless network for users

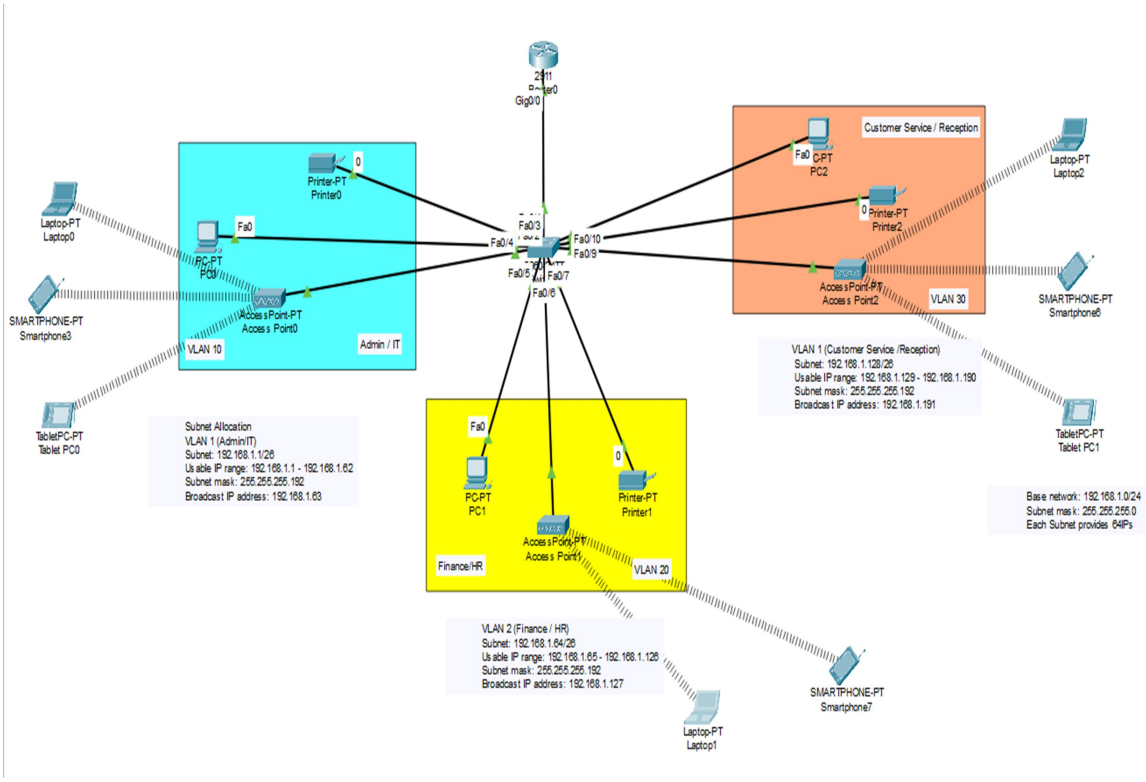
E. Interdepartmental Communications: Host devices in the network are required to obtain IPv4 address automatically

F. Test and Verifications: Devices in all departments are required to communicate with each other

- **Networking Commands Used**

Sr no	Networking Commands
1	Line vty
2	SSH (Transport)
3	Vlan
4	Switchport (Access , Trunk)
5	DHCP

- **Bolzano Branch**



- **Devices Used**

Sr no	Devices	Qty
1	Router 2911	1
2	Switch	1
3	PC	3
4	Printer	3
5	Access Point	3
6	Laptop	3
7	Smart Phone	3
8	Tablet	2

- **Network Allocations**

Base network: 192.168.1.0/24
Subnet mask: 255.255.255.0
Each Subnet provides 64IPs

Subnet Allocation

VLAN 1 (Admin/IT)	**VLAN 2 (Finance / HR)**	**VLAN 3 (Customer Service /Reception)**
Subnet: 192.168.1.1/26	Subnet: 192.168.1.64/26	Subnet: 192.168.1.128/26
Usable IP range: 192.168.1.1 - 192.168.1.62	Usable IP range: 192.168.1.65 - 192.168.1.126	Usable IP range: 192.168.1.129 - 192.168.1.190
Subnet mask: 255.255.255.192	Subnet mask: 255.255.255.192	Subnet mask: 255.255.255.192
Broadcast IP address: 192.168.1.63	Broadcast IP address: 192.168.1.127	Broadcast IP address: 192.168.1.191

- **DHCP IP Address after Router Configuration**

Details	PC 0	PC 1	PC 2	Printer 0	Printer 1	Printer 2
IP address	192.168.1.2	192.168.1.66	192.168.1.130	192.168.1.6	192.168.1.69	192.168.1.134
GW	192.168.1.1	192.168.1.65	192.168.1.129	192.168.1.1	192.168.1.65	192.168.1.129
Subnet Mask	255.255.255.192	255.255.255.192	255.255.255.192	255.255.255.192	255.255.255.192	255.255.255.192
DNS IP	192.168.1.1	192.168.1.65	192.168.1.129	192.168.1.1	192.168.1.65	192.168.1.129

Details (connectivity wireless)	Laptop	Laptop	Laptop	Smart Phone	Smart Phone	Smart Phone	Tablet	Tablet
IP address	192.168.1.3	192.168.1.67	192.168.1.133	192.168.1.4	192.168.1.68	192.168.1.131	192.168.1.5	192.168.1.132
GW	192.168.1.1	192.168.1.65	192.168.1.129	192.168.1.1	192.168.1.65	192.168.1.129	192.168.1.1	192.168.1.129
Subnet Mask	255.255.255.192	255.255.255.192	255.255.255.192	255.255.255.192	255.255.255.192	255.255.255.192	255.255.255.192	255.255.255.192
DNS IP	192.168.1.1	192.168.1.65	192.168.1.129	192.168.1.1	192.168.1.65	192.168.1.129	192.168.1.1	192.168.1.129

Details (connectivity wireless)	Tablet	Tablet
IP address	192.168.1.5	192.168.1.132
GW	192.168.1.1	192.168.1.129
Subnet Mask	255.255.255.192	255.255.255.192
DNS IP	192.168.1.1	192.168.1.129

- **Switch Configuration (In details)**

Configuration	Commands	Description
Switch(config)#	hostname HQ-SW	
HQ-SW(config)#	banner motd #This is HQ switch. Unauthorized access is strictly prohibited#	
HQ-SW(config)#	line console 0	
HQ-SW(config-line)#	password cisco	
HQ-SW(config-line)#	login	
HQ-SW(config-line)#	exec-timeout 3 0	
HQ-SW(config-line)#	logging synchronous	
HQ-SW(config-line)#	exit	
HQ-SW(config)#	enable password cisco	
HQ-SW(config)#	line vty 0 15	(Tells a Cisco device to configure all 16 virtual terminal (VTY) lines—numbered 0 through 15 —which are used for remote access via telnet or SSH)
HQ-SW(config-line)#	password cisco	
HQ-SW(config-line)#	login	
HQ-SW(config-line)#	exit	
HQ-SW(config)#	service password-encryption	
HQ-SW(config)#	username cisco password cisco	
HQ-SW(config)#	ip domain-name cisco.com	
HQ-SW(config)#	crypto key generate rsa general-key modulus 1024	The name for the keys will be: HQ-SW.cisco.com % The key modulus size is 1024 bits % Generating 1024 bit RSA keys, keys will be non-exportable...[OK] *Mar 1 0:22:18.444: %SSH-5-ENABLED: SSH 1.99 has been enabled The name for the keys will be: HQ-SW.cisco.com
HQ-SW(config)#	ip ssh version 2	
HQ-SW(config)#	line vty 0 15	
HQ-SW(config-line)#	login local	
HQ-SW(config-line)#	transport input ssh	
HQ-SW(config-line)#	exit	
HQ-SW(config)#	access-list 1 permit 192.168.1.0 0.0.0.255	

HQ-SW(config)#	access-list 1 deny any	
HQ-SW(config)#	line vty 0 15	
HQ-SW(config-line)#	access-class 1 in	
HQ-SW(config-line)#	do wr	Building configuration… HQ-SW con0 is now available
Press RETURN to get started - Message displayed (This is HQ switch. Unautorised access is strictly prohibited) you nee to enter the password to login in switch		
HQ-SW(config)#	interface range fastEthernet 0/2-4	global configuration command that lets you configure multiple interfaces at once
HQ-SW(config-if-range)#	switchport mode access	command configures a switch port to operate exclusively as an access port, carrying traffic for a single VLAN and disabling trunking
HQ-SW(config-if-range)#	exit	
HQ-SW(config)#	vlan 10	
HQ-SW(config-vlan)#	name admin	
HQ-SW(config-vlan)#	vlan 20	
HQ-SW(config-vlan)#	name hr	
HQ-SW(config-vlan)#	vlan 30	
HQ-SW(config-vlan)#	name cs	
HQ-SW(config-vlan)#	exit	
HQ-SW(config)#	interface range fastEthernet 0/2-4	global configuration command that lets you configure multiple interfaces at once
HQ-SW(config-if-range)#	switchport mode access	
HQ-SW(config-if-range)#	switchport access vlan 10	
HQ-SW(config-if-range)#	interface range fastEthernet 0/5-7	global configuration command that lets you configure multiple interfaces at once
HQ-SW(config-if-range)#	switchport mode access	
HQ-SW(config-if-range)#	switchport access vlan 20	
HQ-SW(config-if-range)#	interface range fastEthernet 0/8-10	global configuration command that lets you configure multiple interfaces at once
HQ-SW(config-if-range)#	switchport mode access	
HQ-SW(config-if-range)#	switchport access vlan 30	
HQ-SW(config-if-range)#	exit	
HQ-SW(config)#	fastEthernet 0/1	

HQ-SW(config)#	switchport mode trunk	is a Cisco IOS interface-configuration command that forces a switch port into permanent 802.1Q trunking mode, allowing it to carry multiple VLANs across a single link.
HQ-SW(config-line)#	do wr	Building configuration… HQ-SW is now configured

- **Router Configuration (In detail)**

Configuration	Commands	Description
Router(config)#	interface gig0/0	
Router(config-if)#	no shutdown	%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/0, changed state to up
Router(config-if)#	exit	
Router(config)#	interface gig0/0.10	%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/0.10, changed state to up
Router(config-subif)#	encapsulation dot1Q 10	
Router(config-subif)#	ip address 192.168.1.1 255.255.255.192	
Router(config-subif)#	interface gig0/0.20	%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/0.20, changed state to up
Router(config-subif)#	encapsulation dot1Q 20	
Router(config-subif)#	ip address 192.168.1.65 255.255.255.192	
Router(config-subif)#	interface gig0/0.30	%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/0.30, changed state to up
Router(config-subif)#	encapsulation dot1Q 30	
Router(config-subif)#	ip address 192.168.1.129 255.255.255.192	
Router(config-subif)#	do wr	Building configuration
Router(config-subif)#	exit	
Router(config)#	service dhcp	
Router(config)#	ip dhcp pool admin	
Router(dhcp-config)#	network 192.168.1.0 255.255.255.192	
Router(dhcp-config)#	default-router 192.168.1.1	
Router(dhcp-config)#	dns-server 192.168.1.1	
Router(dhcp-config)#	exit	
Router(config)#	ip dhcp pool hr	
Router(dhcp-config)#	network 192.168.1.64 255.255.255.192	

Router(dhcp-config)#	default-router 192.168.1.65	
Router(dhcp-config)#	dns-server 192.168.1.65	
Router(dhcp-config)#	exit	
Router(config)#	ip dhcp pool CS	
Router(dhcp-config)#	network 192.168.1.128 255.255.255.192	
Router(dhcp-config)#	default-router 192.168.1.129	
Router(dhcp-config)#	dns-server 192.168.1.129	
Router(dhcp-config)#	do wr	Building configuration

Ping Status

Cisco Packet Tracer PC Command Line 1.0
C:\>ping 192.168.1.130

Pinging 192.168.1.130 with 32 bytes of data:

Request timed out.
Reply from 192.168.1.130: bytes=32 time=1ms TTL=127
Reply from 192.168.1.130: bytes=32 time=1ms TTL=127
Reply from 192.168.1.130: bytes=32 time<1ms TTL=127

Ping statistics for 192.168.1.130:
Packets: Sent = 4, Received = 3, Lost = 1 (25% loss),
Approximate round trip times in milli-seconds:
Minimum = 0ms, Maximum = 1ms, Average = 0ms

C:\>ping 192.168.1.66

Pinging 192.168.1.66 with 32 bytes of data:

Request timed out.
Reply from 192.168.1.66: bytes=32 time<1ms TTL=127
Reply from 192.168.1.66: bytes=32 time=1ms TTL=127
Reply from 192.168.1.66: bytes=32 time<1ms TTL=127

Ping statistics for 192.168.1.66:
Packets: Sent = 4, Received = 3, Lost = 1 (25% loss),
Approximate round trip times in milli-seconds:
Minimum = 0ms, Maximum = 1ms, Average = 0ms

Final Output

We have successfully designed a standalone system for ABC Company as per their Requirement

Thank You……